WORKED SOLUTION

STEP III 2023, Question 9

Mechanics • Two-Particle String Motion

STEP III 2023, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

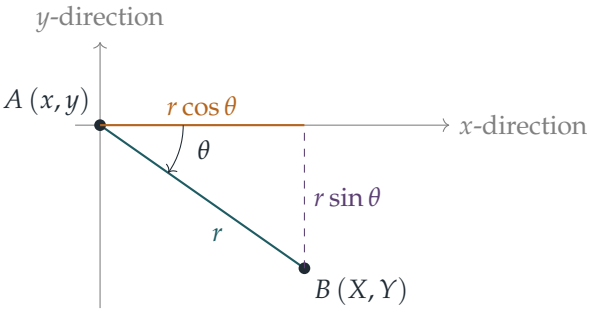
Describe the system by the position of A and the direction θ of the string. The tension acts only along the string, so each particle's acceleration has no component perpendicular to it. Comparing the two particles' perpendicular accelerations then forces $\ddot{\theta} = 0$: the string turns at a constant rate. After that, conservation of momentum finishes the job.

Solution

Throughout, the particles move on a smooth horizontal plane, the string is light and inextensible, and (as we may assume) it stays taut, so $AB = r$ at all times.

Part (i)

The vector $\overrightarrow{AB}$ has length r. Since θ is measured *clockwise* from the positive x-axis, the vector points below the x-direction when $0 < \theta < \frac{\pi}{2}$, as in the diagram.



Going from A to B we move $r\cos\theta$ in the x-direction and $r\sin\theta$ in the *negative* y-direction, so

$$X = x + r\cos\theta, \qquad Y = y - r\sin\theta.$$

The same expressions hold for every θ, because $(\cos\theta, -\sin\theta)$ is the unit vector at angle θ clockwise from the x-axis.

Part (ii)

Differentiating with respect to t (with r constant, and θ a function of t):

$$\dot{X} = \dot{x} - r\dot{\theta}\sin\theta, \qquad \dot{Y} = \dot{y} - r\dot{\theta}\cos\theta,$$
$$\ddot{X} = \ddot{x} - r\ddot{\theta}\sin\theta - r\dot{\theta}^2\cos\theta, \qquad \ddot{Y} = \ddot{y} - r\ddot{\theta}\cos\theta + r\dot{\theta}^2\sin\theta.$$

Part (iii)

The tension T pulls A towards B, along the unit vector $(\cos\theta, -\sin\theta)$, and pulls B towards A, along $(-\cos\theta, \sin\theta)$. By Newton's second law,

$$m\ddot{x} = T\cos\theta, \quad m\ddot{y} = -T\sin\theta, \qquad M\ddot{X} = -T\cos\theta, \quad M\ddot{Y} = T\sin\theta.$$

Hence

$$m(\ddot{x}\sin\theta + \ddot{y}\cos\theta) = T\cos\theta\sin\theta - T\sin\theta\cos\theta = 0,$$

and in exactly the same way $M(\ddot{X}\sin\theta + \ddot{Y}\cos\theta) = 0$. Dividing by m and M gives both required results. (Geometrically: $(\sin\theta, \cos\theta)$ is perpendicular to the string, and neither particle accelerates in that direction.)

> **Ask yourself** We have two equations saying "no perpendicular acceleration". How do they tell us anything about θ?

Subtract the first from the second and substitute the expressions from part (ii) for $\ddot{X} - \ddot{x}$ and $\ddot{Y} - \ddot{y}$:

$$\begin{aligned} 0 &= (\ddot{X} - \ddot{x})\sin\theta + (\ddot{Y} - \ddot{y})\cos\theta \\ &= (-r\ddot{\theta}\sin\theta - r\dot{\theta}^2\cos\theta)\sin\theta + (-r\ddot{\theta}\cos\theta + r\dot{\theta}^2\sin\theta)\cos\theta \\ &= -r\ddot{\theta}\,(\sin^2\theta + \cos^2\theta) = -r\ddot{\theta}. \end{aligned}$$

Since $r \neq 0$, $\ddot{\theta} = 0$, so $\dot{\theta}$ is constant.

To find the constant, use the initial conditions. At $t = 0$, B is at $(r, 0)$ relative to A, so $\theta(0) = 0$. Also $\dot{y}(0) = u$ and $\dot{Y}(0) = 0$, so the expression for $\dot{Y}$ in part (ii) gives

$$0 = u - r\dot{\theta}(0)\cos 0 \quad \Longrightarrow \quad \dot{\theta}(0) = \frac{u}{r}.$$

(The $\dot{X}$ equation is consistent: $0 = 0 - r\dot{\theta}(0)\sin 0$.) So $\dot{\theta} = u/r$ for all t, and integrating with $\theta(0) = 0$,

$$\theta = \frac{ut}{r},$$

as required.

Part (iv)

Adding the equations of motion from part (iii) in pairs,

$$m\ddot{x} + M\ddot{X} = T\cos\theta - T\cos\theta = 0, \qquad m\ddot{y} + M\ddot{Y} = -T\sin\theta + T\sin\theta = 0.$$

This is conservation of momentum: the tension is internal to the system. Integrating the second equation once, $m\dot{y} + M\dot{Y}$ is constant, equal to its initial value $mu + M \cdot 0 = mu$. Integrating again, with $y(0) = Y(0) = 0$,

$$my + MY = mut.$$

Part (v)

Substitute $Y = y - r\sin\theta$ from part (i) and $\theta = ut/r$ from part (iii):

$$my + M\Big(y - r\sin\frac{ut}{r}\Big) = mut \quad \Longrightarrow \quad y = \frac{1}{m+M}\Big(mut + Mr\sin\frac{ut}{r}\Big),$$

which is the required result.

Part (vi)

Differentiating the result of part (v),

$$\dot{y} = \frac{1}{m+M}\Big(mu + Mu\cos\frac{ut}{r}\Big) = \frac{u}{m+M}\Big(m + M\cos\frac{ut}{r}\Big).$$

At the time $t = \pi r/u$, when the string has turned through half a revolution, $\cos(ut/r) = -1$ and

$$\dot{y} = \frac{u(m-M)}{m+M},$$

which is negative when $M > m$ (as $u > 0$). So A's velocity has a negative y component at that moment, and by continuity on an interval of times around it.

Checks

- At $t = 0$ the formula in part (v) gives $\dot{y} = u(m+M)/(m+M) = u$, matching the initial velocity of A.
- The same method in the x-direction gives $mx + MX = Mr$, so $x = \dfrac{Mr}{m+M}(1 - \cos\theta)$. Together with part (v), the centre of mass moves in a straight line at speed $mu/(m+M)$ in the y-direction while both particles circle it at angular speed u/r, which is the familiar picture. The tension is then the constant $\dfrac{mM}{m+M}\dfrac{u^2}{r} > 0$, so the string really does stay taut.
- If $M \to \infty$, B stays fixed and part (v) gives $y = r\sin(ut/r)$: A moves in a circle of radius r about B, as it should.

- With $m = 1$, $M = 3$, $u = r = 1$, a computer simulation of the two particles joined by a very stiff spring gave $y \approx 0.7854$ and $\dot{y} \approx -0.5000$ at $t = \pi$, matching $\frac{\pi}{4}$ and $\frac{1-3}{1+3} = -\frac{1}{2}$.

Common slips

- Sign errors from the clockwise convention: $Y = y - r\sin\theta$, so $\dot{Y}$ has $-r\dot{\theta}\cos\theta$, not $+$.
- Forgetting to justify the constant u/r: it comes from $\dot{Y}(0) = 0$, not from A's speed alone.
- In part (vi), claiming $\dot{y} < 0$ "eventually" without exhibiting a time; $t = \pi r/u$ gives a definite one.

Beyond STEP

This is the two-body problem in its simplest form: the motion splits into uniform motion of the centre of mass plus motion relative to it, governed by the reduced mass $mM/(m+M)$. The same splitting underlies planetary orbits and the vibrations of diatomic molecules. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops centre-of-mass coordinates and reduced mass carefully.

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